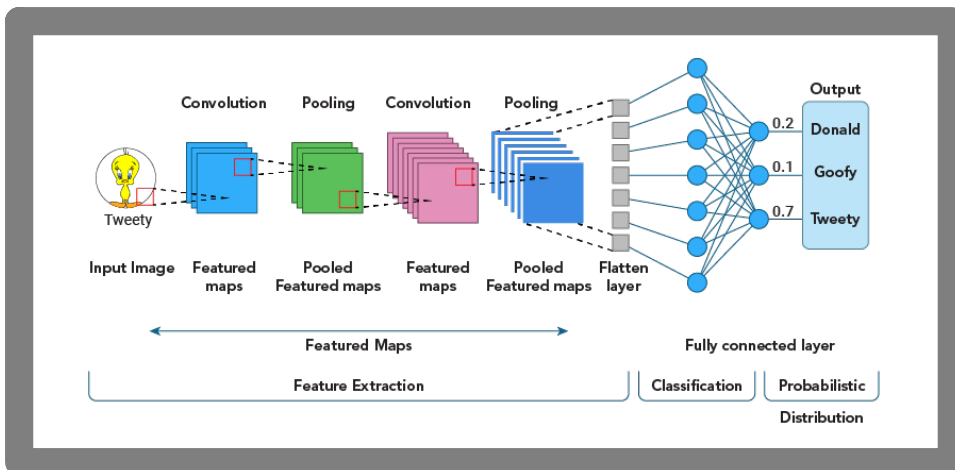




A CNN is a deep learning model designed to process grid-structured data [eg. images]. It consists of three primary stages :-

Feature Extraction [Convolution, Activation and Pooling layers]

Classification [Flattening and Fully connected Layer].

① Convolutional Layer [CONV] - This is the core building block of a CNN, responsible for detecting features like edges, colours, textures etc.

How it works → Small matrix of weights called kernels (or filters) slide across the input image. At every position, the network computes the dot product between the filter and the underlying pixels.

Hyperparameters →

- ① Filter Size - Common dimension like 3×3 , 5×5 , 7×7
- ② Stride - The number of pixels the filter shifts per step [eg. stride 1 or 2]
- ③ Padding - Adding extra rows/columns of

zeros around the image to prevent data loss at the border and retain the spatial size.

Output A feature map that highlights where the specific patterns were detected in the image

② Activation Layer [usually ReLU] — Convolution is a linear process, but real-world data is highly complex and non-linear.

How it works → An activation function, primarily ReLU [rectified linear unit], is applied element-wise to feature map. The formula is $f(x) = \max(0, x)$ which converts all negative pixel value to zero.

Purpose → This introduces non-linearity, allowing the network to learn complex, hierarchical features without slowing down the training process.

③ Pooling Layer [pool] — Inserted periodically between convolutional layers performs downsampling to reduce the spatial dimensions (width and height) of the feature map.

How it works → A filter [like 2×2 , 3×3 with stride 2] slides across the feature map and summarizes the values. Common types :-

① Max Pooling — Takes the maximum value in the filter's view.

② Average Pooling — Takes the average of all values in the filter's view.

Purpose → Decreases computational cost, reduces the memory usage, controls overfitting, and makes the model invariant to the minor shifts and distortions in the image.

④ Flattening Layer → Before entering the final classification phase, the multi-dimensional feature maps from the previous layers must be reshaped.

How it works → It takes 3D feature maps and unrolls them into a long, continuous 1D column vector.

Purpose → Converts the spatial data into the standard format accepted by traditional artificial neural network.